Diagnostics and Service System

High-Level Design (HLD)

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This document describes the software architecture of the diagnostics and service domain: UDS service handling, fault memory management, routine control and over-the-air update components and interfaces.

1 Revision history

Revision	Date	Author	Description
B	2026-05-14	A. Kumar	Initial architecture baseline

2 Scope and context

2.1 Scope

The diagnostics domain covers UDS service dispatch, DTC storage and retrieval, routine execution and secure over-the-air update staging. Out of scope: tester application design and production flash programming hardware details.

2.2 System context

The domain runs on DiagnosticsEcu connected to all other ECUs through the central gateway over DoIP and CAN.

3 Software architecture

3.1 Component breakdown

Component	Type	Description
UdsServiceDispatcher	ApplicationSwComponentType	UDS service request routing
FaultMemoryManager	ApplicationSwComponentType	DTC storage and aging
RoutineControlManager	ApplicationSwComponentType	Diagnostic routine execution
OtaUpdateStager	ApplicationSwComponentType	Over-the-air package staging
SecurityAccessManager	ApplicationSwComponentType	UDS security access levels
DiagSignalService	ServiceSwComponentType	Diagnostics signal service
DiagGatewayAbstraction	EcuAbstractionSwComponentType	Diagnostic gateway abstraction
DiagnosticsMainComposition	CompositionSwComponentType	Composition of the diagnostics com

3.2 Ports and interfaces

Port	Direction	Interface	Component
UdsRequestIn	required	UdsRequestI	UdsServiceDispatcher
DtcStatusOut	provided	DtcStatusI	FaultMemoryManager
RoutineResultOut	provided	RoutineResultI	RoutineControlManager
OtaProgressOut	provided	OtaProgressI	OtaUpdateStager
SecurityTokenOut	provided	SecurityTokenI	SecurityAccessManager
DiagGatewayIn	required	DiagSignalSetI	DiagSignalService
GatewayToTester	provided	DiagRoutingI	DiagGatewayAbstraction
FaultSnapshotOut	provided	FaultSnapshotI	FaultMemoryManager

3.3 Signal inventory

Signal	Interface	Sender	Receiver	Bits
UdsResponseCode	UdsRequestI	UdsServiceDispatcher	DiagGatewayAbstractio	8
ActiveDtcCount	DtcStatusI	FaultMemoryManager	UdsServiceDispatcher	8
RoutineStatus	RoutineResultI	RoutineControlManager	FaultMemoryManager	4
UpdateProgress	OtaProgressI	OtaUpdateStager	UdsServiceDispatcher	8
SecurityLevel	SecurityTokenI	SecurityAccessManager	OtaUpdateStager	4
FaultSnapshotData	FaultSnapshotI	FaultMemoryManager	RoutineControlManager	32
TesterPresent	TesterPresentI	DiagGatewayAbstraction	UdsServiceDispatcher	1

4 Functional flows

4.1 UDS service flow

UdsServiceDispatcher consumes UdsRequestIn from the tester through GatewayToTester, validates SecurityLevel via SecurityAccessManager and dispatches to FaultMemoryManager or RoutineControlManager as required. UdsResponseCode is returned to the tester.

4.2 Fault memory flow

FaultMemoryManager stores confirmed DTCs with FaultSnapshotData and ages them over drive cycles. ActiveDtcCount is published to UdsServiceDispatcher for ReadDTCInformation services.

4.3 OTA staging flow

OtaUpdateStager receives a signed update package, validates it with SecurityAccessManager and stages it in the standby flash bank. UpdateProgress is reported to the tester until activation completes.

5 Timing and runnables

TimingEvent UdsDispatchEvent triggers DispatchUdsRequestRunnable every 10 ms. TimingEvent FaultAgingEvent triggers AgeFaultMemoryRunnable every 1000 ms. TimingEvent OtaVerifyEvent triggers VerifyOtaPackageRunnable every 200 ms. All runnables are scheduled by DiagTask on DiagnosticsEcu.

6 Basic software integration

Signal routing uses Com and PduR. Incoming DoIP requests are handled by SoAd and TcpIp. NvM stores the DTC memory and security seeds. Rte provides the port API used by all diagnostics application components.

Mode management is performed by BswM and EcuM. Watchdog supervision for DiagTask is performed by WdgM. Flash access for OTA staging uses Fls through the memory abstraction.

The diagnostic path routes UDS requests through Dcm and Dem. DiagSignalService consumes DiagSignalSetI and forwards session state changes to the network management stack.

7 Diagnostics and configuration

DTCs for diagnostic session timeout, security access violation and OTA package verification failure are defined in the diagnostics specification. The HLD defines the internal reporting path: component to Dcm to SoAd.

Configuration parameters are grouped into the domain configuration container and written through NvM during shutdown.

8 Known limitations

OtaUpdateStager supports a single standby flash bank in this revision; dual-bank seamless update is planned for a later revision.

9 Open questions

Open: calibration ownership for cross-domain interfaces and the timeline for production software freeze are tracked outside this document.